How do different trading rules change profit and loss?

First understand trades and the account, then compare entry conditions, holding periods, and when to take profit or stop a loss.

Reading order	Questions to answer
Issue 1-4: Decisions and execution	When is the signal known? When does the stock holding actually begin?
Issue 5-8: Account profit and loss	How much is invested? How many shares are held? How much is earned?
Issue 9-14: Entry and holding conditions	What changes with fees, entry prices, lookback periods and thresholds?
Issue 15-22: Exit rules	How do fixed holding periods, profit targets, stop losses and tracking a high differ?
Issue 23-25: Re-entry and full-period comparisons	How long must you wait after selling? How do trade counts and fees affect the result?
Issue 26: Stop losses and the number of shares bought	For a planned loss of 20 USD, how many shares should you buy and how much cash should you retain?

First compare exit rules using the same 1/18 entry and 3/28 valuation date; compare full-period holding conditions separately over 1/2-6/28. Do not rank results from different periods together. Part D uses OLS forecasts for decisions. Optional FinTech extensions outside this four-part course: Week 6 examines trading costs and Week 7 evaluates risk.

How do closing prices determine whether to hold stock or cash?

Compare today's close with the previous trading day's close: a rise requests stock; otherwise, cash.

Item	Setting
Data	AAPL, 2024/1/2 to 6/28, 124 trading days
Initial capital	1,000 USD cash; no additional funding
Execution	Decide after the close; trade at the next trading day's close
Holding stock	All stock or all cash; fractional shares allowed; no borrowing or short selling
Base account	No dividends, fees, tax or cash interest
Final day	Valuation only; no new trades

Closing prices come from the market; execution at the next close is an assumption of this example.

Predict: 1/8 exceeds 1/5, while 1/9 is below 1/8. Does each day request stock or cash?

```
import numpy as np
import pandas as pd
import yfinance as yf
aapl = yf.Ticker("AAPL").history(
    start="2024-01-02", end="2024-07-01",
    interval="1d", auto_adjust=False, actions=True)
aapl.index = aapl.index.strftime("%Y-%m-%d")
close = aapl["Close"]
dates = ["2024-01-05", "2024-01-08", "2024-01-09", "2024-01-10"]
pd.DataFrame({'Observation date': dates,
    'Closing price (USD/share)': [f"{close[d]:.2f}" for d in dates]})
```

Observation date	Closing price (USD/share)
2024-01-05	181.18
2024-01-08	185.56
2024-01-09	185.14
2024-01-10	186.19

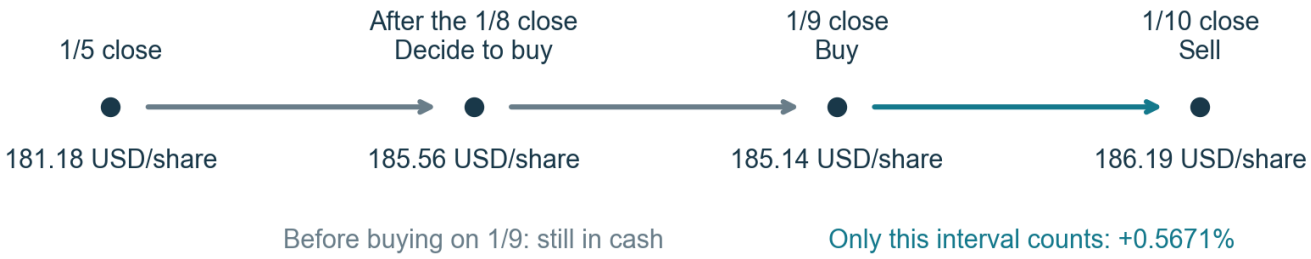
1/8: 185.56 exceeds 181.18, requesting stock; 1/9: 185.14 is below 185.56, requesting cash. The request forms after the close, with execution at the next trading day's close. Issue 1 tracks the first trade through these four dates.

ISSUE 01

After deciding to buy on 1/8, when does profit or loss begin?

Establish the entry time before selecting the two prices used for profit and loss.

AAPL 2024: buy on 1/9, sell on 1/10



The holding period is 1/9 to 1/10; the earlier 1/5 to 1/8 rise does not belong to this trade.

The 1/9 close of 185.14 is below 185.56 on 1/8, so the after-close request changes to cash and the sale occurs at the 1/10 close. Stock is still held until the sale; do not omit that interval's profit or loss.

$$\frac{186.19}{185.14} - 1 \approx +0.5671\%$$

This interval earns about 0.5671%. It is a price return under assumed execution at the closes, not a guarantee that actual orders can execute at those prices.

Can you buy in after-hours trading that day?

U.S. stocks can trade after hours, generally from 16:00 to 20:00 Eastern Time; access depends on the broker. But the 1/8 after-hours price is not the 1/8 closing price of 185.56; these daily data contain no after-hours execution prices. This example specifies buying at the 1/9 close, not that the market requires waiting until the next day.

Discussion

If you instead buy after hours on 1/8, can you directly use 185.56 as the entry price?

Sources: [2], [6], [7].

ISSUE 02

How does a higher close become a request to hold stock?

Calculate the 1-day return from two closing prices, then turn it into an explicit rule.

C is the closing price (USD/share); t indexes trading days and t-1 is the previous trading day. r is the return between those two closes.

$$r_t = \frac{C_t}{C_{t-1}} - 1$$

$$r_{\text{Jan8}} = \frac{185.56}{181.18} - 1 \approx +2.4175\%$$

1/5 is Friday and the next trading day is 1/8. signal=1 requests stock and =0 requests cash. The 1/8 rise requests stock, but buying occurs at the 1/9 close.

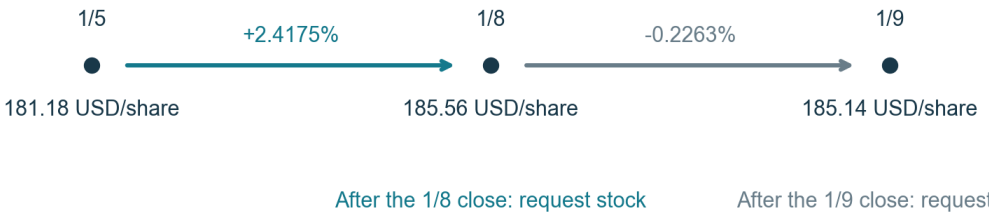
Predict: Do 1/8 and 1/9 each request stock or cash?

```
asset_return = close.pct_change(fill_method=None)
signal = (asset_return > 0).astype(float).where(asset_return.notna())
dates = ["2024-01-08", "2024-01-09"]
pd.DataFrame({
    'Observation date': dates,
    'Previous close': [f"{close.shift(1)[d]:.2f}" for d in dates],
    'Current close': [f"{close[d]:.2f}" for d in dates],
    '1-day return': [f"{asset_return[d]:.4%}" for d in dates],
    'Request': ['Holding stock' if signal[d] == 1 else 'Cash' for d in dates]})
```

Observation date	Previous close	Current close	1-day return	Request
2024-01-08	181.18	185.56	2.4175%	Holding stock
2024-01-09	185.56	185.14	-0.2263%	Cash

1/2 lacks a previous close: nan means the request cannot yet be calculated, not 0. A return >0 requests stock; otherwise, cash. It does not guarantee a further rise.

AAPL 2024/1/5-1/9: changes between adjacent closes



+2.4175% requests stock; -0.2263% requests cash. A request is not a trade already completed that day.

Only the discussion changes the lookback: on 1/11, the 5-day calculation uses 181.91 on 1/4, while the 1-day calculation uses 186.19 on 1/10. Both end at 185.59 USD/share on 1/11.

Discussion

If the 1/11 lookback changes to a 5-day return, will the request match the 1-day rule?

Sources: [4].

ISSUE 03

Why is there no account loss on 1/9 when buying at that close?

A stock price fall does not mean a cash account loses money; check whether stock was held during that interval.

The 1/8 after-close request leads to buying at the 1/9 close; the first holding-period profit or loss ends on 1/10. From 1/8 to 9, the price falls from 185.56 to 185.14 USD/share, while the account remains 1,000 USD cash.

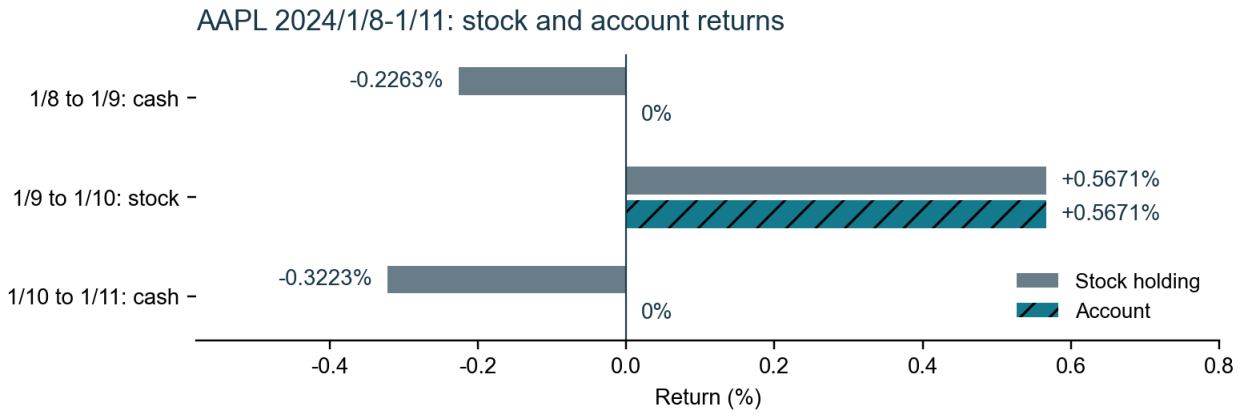
The 1/10 row covers the 1/9 close to the 1/10 close. Its holding is determined by the 1/8 request, two rows earlier, hence signal.shift(2). This shift applies to the next-close execution used here.

position records whether stock is held at the interval's start. after_close records the holding after close execution, determined by the previous trading day's request (1 stock, 0 cash). account_return uses the stock return when held and 0 for cash. wealth is the account value in USD, compounding account returns from the initial 1,000.

Predict: On which day are stock and account returns equal and nonzero (not 0)?

```
# Initial unavailable requests are nan; fillna(0) treats them as cash
position = signal.shift(2).fillna(0)
after_close = signal.shift(1).fillna(0)
after_close.iloc[-1] = position.iloc[-1]  # Final day: valuation only, no new trades
account_return = asset_return.where(position.eq(1), 0.0)
wealth = 1000 * (1 + account_return).cumprod()
dates = ["2024-01-09", "2024-01-10", "2024-01-11"]
pd.DataFrame({
    'Return end date': dates,
    'Held from previous close': ['Held' if position[d] == 1 else 'Not held' for d in dates],
    'Stock return': [f"{asset_return[d]:.4%}" for d in dates],
    'Account return': [f"{account_return[d]:.4%}" for d in dates]})
```

Return end date	Held from previous close	Stock return	Account return
2024-01-09	Not held	-0.2263%	0.0000%
2024-01-10	Held	0.5671%	0.5671%
2024-01-11	Not held	-0.3223%	0.0000%



Stock is held from 1/9 to 10: the account earns 0.5671% and has 1,005.67 USD after selling. Cash is held from 1/10 to 11, so the stock's 0.3223% fall does not change the account value.

Discussion

If the 1/8 to 1/9 stock loss is wrongly included, what would the 1/9 account's 1,000 USD be calculated as?

Sources: [3], [4].

ISSUE 04

Does a continuing stock request mean buying again every day?

When the stock allocation is unchanged, retain the existing shares; do not pay for another purchase.

The previous trading day's after-close request determines the action at this close: buy or sell only if the pre-close position differs; otherwise, do not trade. changes is after_close minus position: 1 buys, -1 sells, and 0 does not trade. Track 1/19 buying through 1/25 selling; the trading day before 1/22 is 1/19.

Predict: If all shares are still held, should another purchase payment be deducted on 1/22?

```
changes = after_close - position
state = {1: 'Holding stock', 0: 'Cash'}
dates = ["2024-01-19", "2024-01-22", "2024-01-23", "2024-01-24", "2024-01-25"]
pd.DataFrame({
    'Observation date': dates,
    'Pre-close position': [state[position[d]] for d in dates],
    "Previous trading day's request": [state[after_close[d]] for d in dates],
    'Action at this close': ['Buy' if changes[d] > 0 else
        'Sell' if changes[d] < 0 else 'Continue holding' if after_close[d] == 1
        else 'Stay in cash' for d in dates]})
```

Observation date	Pre-close position	Previous trading day's request	Action at this close
2024-01-19	Cash	Holding stock	Buy
2024-01-22	Holding stock	Holding stock	Continue holding
2024-01-23	Holding stock	Holding stock	Continue holding
2024-01-24	Holding stock	Holding stock	Continue holding
2024-01-25	Holding stock	Cash	Sell



1/22, 1/23 and 1/24 only revalue the holding; there is no further purchase payment.

An all-stock account has no spare cash. Treating 'holding stock' as a new purchase each day would require additional funding or unreported borrowing; neither is allowed by this part's rules.

Discussion

If the account is all stock and the previous trading day also requested stock, what trading action is needed at today's close?

Sources: [3].

ISSUE 05

How much does an account earn with half in stock and half in a deposit?

Account profit and loss combine stock profit and loss with deposit interest, not just the stock portion.

From the 2024/1/9 close to 10, AAPL moves from 185.14 to 186.19 USD/share; initial capital is 1,000 USD. There are no fees, dividends or intermediate trades.

Assume a USD deposit earns 1% annually with 365-day simple interest for 1 calendar day; this is not a bank quote. With 500 USD in each asset, first add stock profit and interest:

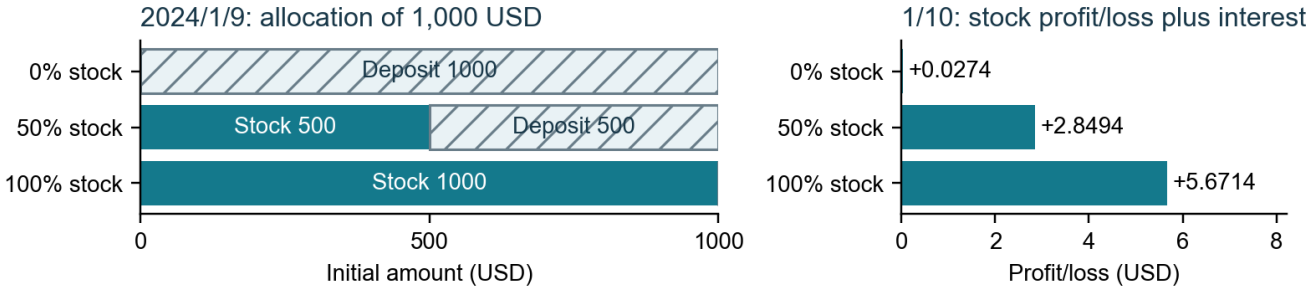
$$500\left(\frac{186.19}{185.14}-1\right)+500\left(\frac{0.01}{365}\right)\approx 2.8357+0.0137=2.8494 \quad \text{(USD)}$$

Total profit divided by 1,000≈0.28494%. w is the initial stock weight (0.5 here); r and i are stock and deposit returns over the same period, as decimals. Account return=w×r+(1-w)×i.

Predict: With everything in the deposit, is this day's profit still 0? All output amounts are in USD.

```
weights = np.array([0.0, 0.5, 1.0])
stock_return = close["2024-01-10"] / close["2024-01-09"] - 1
annual_rate = 0.01  # Assumed annual interest rate, not a bank quote
cash_return = annual_rate * 1 / 365  # 1 calendar day
stock_gain = 1000 * weights * stock_return
cash_interest = 1000 * (1 - weights) * cash_return
end_value = 1000 + stock_gain + cash_interest
pd.DataFrame({
    'Stock weight': [f"{w:.0%}" for w in weights],
    'Stock profit/loss': [f"{v:.4f}" for v in stock_gain],
    'Accrued interest': [f"{v:.4f}" for v in cash_interest],
    'Ending value': [f"{v:.4f}" for v in end_value]})
```

Stock weight	Stock profit/loss	Accrued interest	Ending value
0%	0.0000	0.0274	1000.0274
50%	2.8357	0.0137	1002.8494
100%	5.6714	0.0000	1005.6714



0% stock: accrued interest of 0.0274 USD; 100% stock: profit of 5.6714 USD; 50%: add half of each.

Interest is accrued and may not be credited that day. Whether broker-held cash earns interest depends on the terms. This example does not change the main backtest.

Discussion

At the same daily rate, what interest accrues on 500 USD deposited from the 1/12 close to the 1/16 close (4 calendar days)?

Sources: [9], [10].

ISSUE 06

Reinvest all funds, or cap each purchase at 1,000 USD?

Only the amount purchased changes; signals, execution dates and prices are identical.

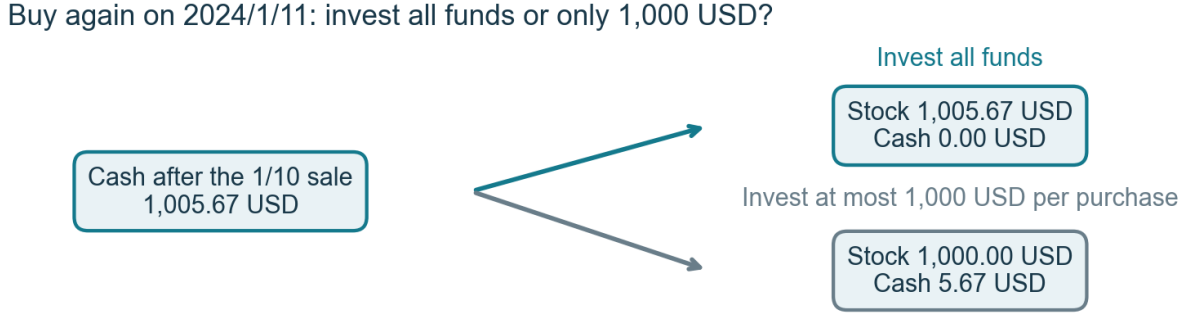
Both start with 1,000 USD: fractional shares allowed; no additions, withdrawals, borrowing, fees or cash interest. Invest all funds or cap each buy at 1,000 USD. Retain the rest as cash; do not resize holdings.

On 1/11, 1,005.67 USD is available. The capped account buys 1,000 and retains 5.67 USD; if its balance is below 1,000, it can buy only with the available funds.

Predict: The accounts have equal value after buying on 1/11 but different share quantities. Are their values still equal on 1/12? All amounts are in USD.

```
cap_cash, cap_shares = 1000.0, 0.0
cap_values, cap_cash_values, cap_amounts = [], [], []
for date, price in close.items():
    amount = 0.0
    if changes[date] > 0:
        amount = min(1000.0, cap_cash)
        cap_shares = amount / price
        cap_cash -= amount
    elif changes[date] < 0:
        cap_cash += cap_shares * price
        cap_shares = 0.0
    cap_values.append(cap_cash + cap_shares * price)
    cap_cash_values.append(cap_cash)
    cap_amounts.append(amount)
cap_wealth = pd.Series(cap_values, index=close.index)
cap_cash_path = pd.Series(cap_cash_values, index=close.index)
dates = ["2024-01-11", "2024-01-12", "2024-06-28"]
pd.DataFrame({'Observation date': dates,
    'All-funds account value': [f"{wealth[d]:.2f}" for d in dates],
    'Value with 1,000 purchase cap': [f"{cap_wealth[d]:.2f}" for d in dates]})
```

Observation date	All-funds account value	Value with 1,000 purchase cap
2024-01-11	1005.67	1005.67
2024-01-12	1007.46	1007.45
2024-06-28	1134.06	1130.73



On 1/11, the all-funds account invests 1,005.67 USD. The 1,000-capped account retains 5.67 USD cash and ends 3.33 USD lower on 6/28.

The difference comes from the capital invested later, not different trade timing. Investing all funds earns more here, but need not be more favorable next time.

Discussion

Only 994.6854 USD cash is available before buying on 2/28. How much stock should the second account buy, in USD?

Sources: [5].

ISSUE 07

Does access to fractional shares always let you invest all 1,000 USD?

Compare whole shares, specified fractional increments and ideal fractional shares; uninvested cash is also an asset.

A fractional share is less than one share. Availability, increments and execution depend on the broker; arbitrary fractional shares are this part's ideal assumption [11].

Buy on 1/9 at 185.14 and sell on 1/10 at 186.19 USD/share, starting with 1,000 USD and no fees. q is shares; B and W are cash and total value (USD); C is the closing price (USD/share).

$$q = \frac{1000}{185.14} \approx 5.40132 \quad \text{(shares)}$$

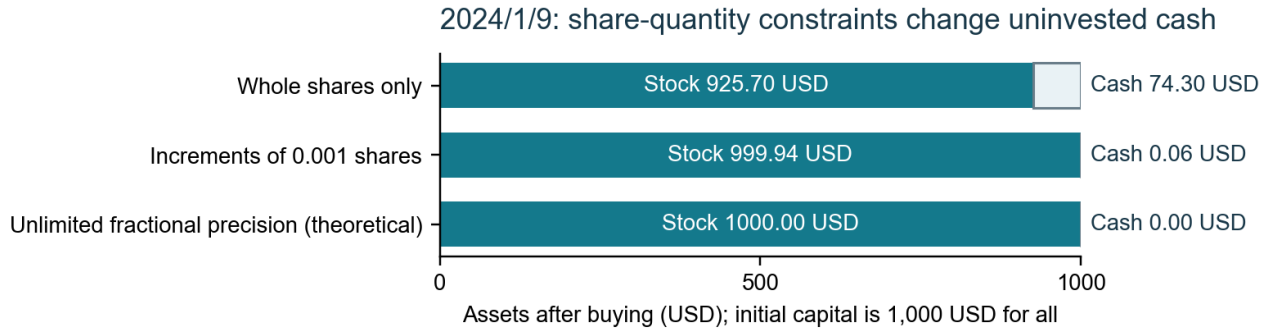
Round 5.40132 shares down to 5; buying 6 requires 6×185.14=1,110.84 USD, exceeding the budget. Add remaining cash B back to total value W:

$$B \approx 1000 - 5(185.14) = 74.30, \quad W = B + qC \approx 74.30 + 5(186.19) \approx 1005.25 \quad \text{(USD)}$$

Predict: Do 0.001-share increments still leave some cash? All table amounts are in USD.

```
entry, exit_price = close["2024-01-09"], close["2024-01-10"]
first_shares = 1000 / entry
quantity_shares = np.array([np.floor(first_shares),
    np.floor(first_shares * 1000) / 1000, first_shares])
quantity_cash = 1000 - quantity_shares * entry
quantity_end = quantity_cash + quantity_shares * exit_price
pd.DataFrame({
    'Constraint': ['Whole shares', '0.001-share increments (assumed)', 'Arbitrary fractions (ideal)'],
    'Shares': [f"{q:.5f}" for q in quantity_shares],
    'Cash after buying': [f"{b:.4f}" for b in quantity_cash],
    'Value after selling': [f"{w:.4f}" for w in quantity_end]})
```

Constraint	Shares	Cash after buying	Value after selling
Whole shares	5.00000	74.3000	1005.2500
0.001-share increments (assumed)	5.40100	0.0589	1005.6711
Arbitrary fractions (ideal)	5.40132	0.0000	1005.6714



Whole shares leave 74.30 USD; 0.001-share increments leave 0.0589 USD; ideal fractional shares invest all funds. Round shares down to avoid overspending.

Discussion

If only 0.01-share increments are accepted, should you buy 5.40 or 5.41 shares? How much cash remains?

Sources: [11].

ISSUE 08

How much does this complete trade earn?

Track the complete account from the 1/19 purchase to the 1/25 sale to identify capital, shares and sale proceeds.

On 1/19, buy with the available 1002.2475 USD at 191.56 USD/share: 1002.2475÷191.56≈5.23203 shares. The quantity stays fixed until selling on 1/25. Table value=shares×closing price (USD); no fees or additional funding.

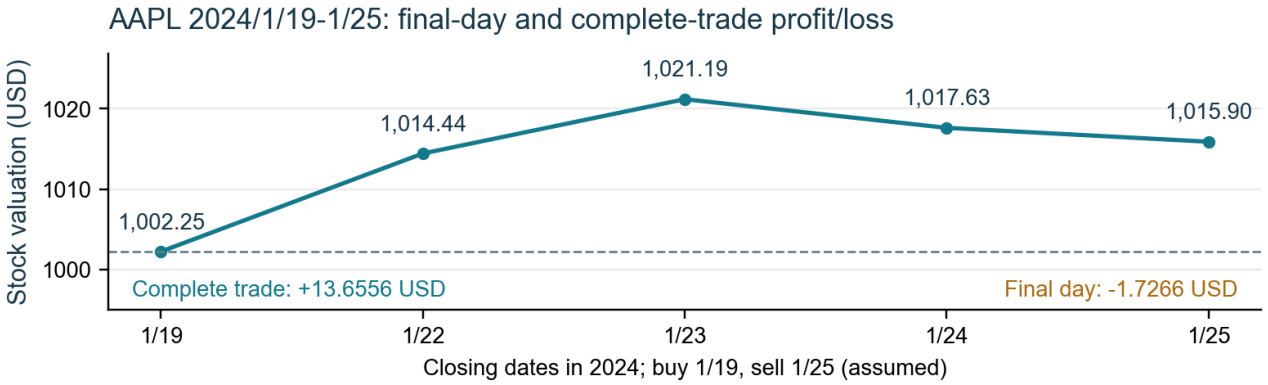
Predict: 1/25 is below 1/24; are assets still worth more than at the 1/19 purchase?

```
trade_dates = ["2024-01-19", "2024-01-22", "2024-01-23",
               "2024-01-24", "2024-01-25"]
trade_start = wealth["2024-01-19"]
trade_shares = trade_start / close["2024-01-19"]
trade_values = trade_shares * close.loc[trade_dates]
ledger = pd.DataFrame({
    'Observation date': trade_dates,
    'Closing price': [f"{close[d]:.2f}" for d in trade_dates],
    'Total value': [f"{v:.4f}" for v in trade_values]})
ledger
```

Observation date	Closing price	Total value
2024-01-19	191.56	1002.2475
2024-01-22	193.89	1014.4381
2024-01-23	195.18	1021.1874
2024-01-24	194.50	1017.6297
2024-01-25	194.17	1015.9031

$$\frac{194.17}{191.56} - 1 \approx +1.3625\%$$

After selling on 1/25, the account has 1015.9031 USD, earning about 13.6556 USD over the complete trade. The final day's fall from 194.50 to 194.17 loses about 1.7266 USD (-0.1697%); that is not the whole trade's profit or loss.



Revalue the same approximately 5.23203 shares at each daily close. The dashed line is the amount invested on 1/19. The three intermediate days only revalue the holding; there are no additional purchases or sales.

Discussion

If shares become 0 after the 1/25 sale, can profit or loss from 1/24 to 1/25 also be counted as 0?

Sources: [4], [5].

ISSUE 09

Does a 0.5671% stock rise still produce a profit after fees?

Keep the 1/9 purchase and 1/10 sale fixed; change only the fee per trade.

Start with 1,000 USD, buying at 185.14 and selling at 186.19 USD/share. Assume a fee of f USD on each purchase and sale: deduct the buy fee before buying shares and the sell fee after selling. Arbitrary fractional shares are allowed and other costs excluded. These fees are not broker quotes.

r is the stock return as a decimal, about 0.0056714; profit is account profit/loss (USD). Invested capital is 1,000-f and stock profit/loss is (1,000-f)r; subtract the two fees for account profit/loss.

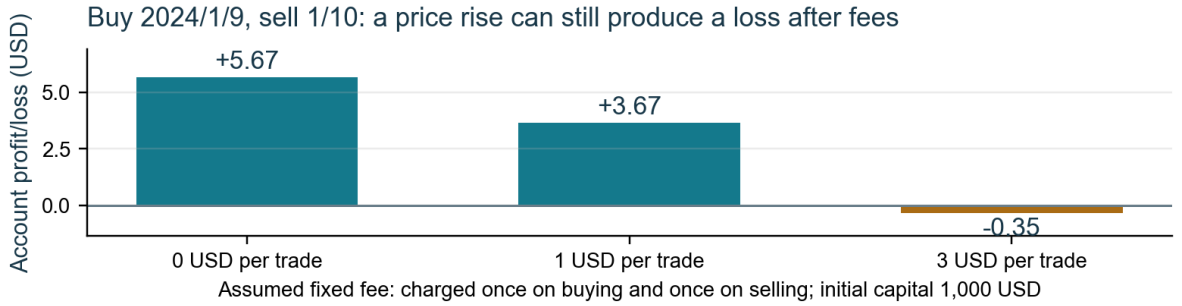
$$\text{profit} = (1000 - f)r - 2f$$

$$f = 3: \quad 997\left(\frac{186.19}{185.14} - 1\right) - 6 \approx -0.35 \quad \text{(USD)}$$

Predict: If buying and selling each cost 3 USD, does a 0.5671% rise cover the 6 USD in fees? All amounts are in USD.

```
entry, exit_price = close["2024-01-09"], close["2024-01-10"]
fees = np.array([0.0, 1.0, 3.0])  # Assumed fee per trade
fee_shares = (1000 - fees) / entry
fee_ends = fee_shares * exit_price - fees
pd.DataFrame({
    'Fee per trade': fees,
    'Shares bought': [f"{q:.5f}" for q in fee_shares],
    'Cash after selling': [f"{w:.2f}" for w in fee_ends],
    'Net profit/loss': [f"{w-1000:+.2f}" for w in fee_ends]})
```

Fee per trade	Shares bought	Cash after selling	Net profit/loss
0.0	5.40132	1005.67	+5.67
1.0	5.39592	1003.67	+3.67
3.0	5.38511	999.65	-0.35



With an assumed fee of 3 USD on buying and selling, total fees are 6 USD. Despite the price rise, this trade loses 0.35 USD in the account.

At 3 USD per trade, the 997 USD stock investment must earn 6 USD to break even: 6÷997≈0.6018%, above this trade's 0.5671%. A price rise may be insufficient to cover fees.

Discussion

What is this trade's net account profit/loss if buying and selling each cost 2 USD?

Sources: [2], [5].

ISSUE 10

Is buying at the next open the same as buying at the next close?

Both rules buy after the decision, but entry prices and exposure periods differ.

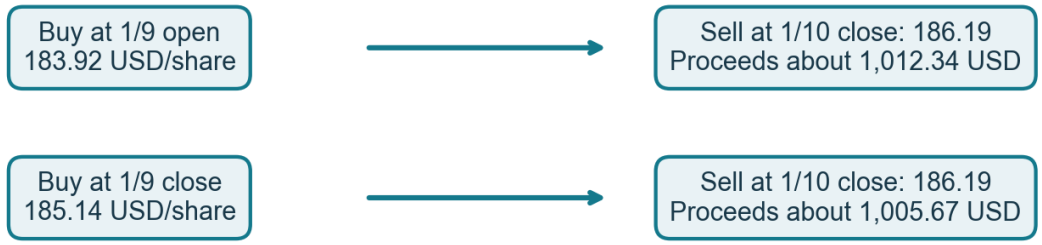
Use the 1/8 after-close buy request and initial 1,000 USD. Both assume selling at the 1/10 close with no fees. Entry changes to the 1/9 open or 1/9 close; both entry price and holding period change together.

Predict: If the open entry price is below the close entry price, which assumption gives higher proceeds at the same sale price?

```
entry_prices = aapl.loc["2024-01-09", ["Open", "Close"]]
exit_price = close["2024-01-10"]
pd.DataFrame({
    'Entry timing': ['1/9 open', '1/9 close'],
    'Entry price (USD/share)': [f"{c:.2f}" for c in entry_prices],
    'Sale price (USD/share)': [f"{exit_price:.2f}"] * 2,
    'Return': [f"{exit_price/c-1:.4%}" for c in entry_prices],
    'Sale proceeds (USD)': [f"{1000*exit_price/c:.2f}" for c in entry_prices]})
```

Entry timing	Entry price (USD/share)	Sale price (USD/share)	Return	Sale proceeds (USD)
1/9 open	183.92	186.19	1.2342%	1012.34
1/9 close	185.14	186.19	0.5671%	1005.67

AAPL buy 2024/1/9, sell 1/10: change only entry timing



Both accounts start with 1,000 USD. Buying at the open includes the 1/9 intraday change; buying at the close does not.

$$\frac{186.19}{183.92} - 1 \approx +1.2342\%$$

The open of 183.92 is below the close of 185.14 USD/share. Here, open-entry proceeds are about 1012.34 USD, above 1,005.67 USD for close entry. This follows from this trade's prices, not that open entry is more favorable on every date.

Buying at the open adds exposure from the 1/9 open to close, but not from the 1/8 close to 1/9 open: the 1/9 open of 183.92 is not the 1/8 close of 185.56 USD/share. Use the open as the starting price; do not change execution to the open while retaining a close-to-close formula.

Discussion

If buying at the 1/9 open instead, should the first profit/loss interval start from Open or 1/8 Close?

Sources: [1], [2].

ISSUE 11

How do final values differ between buy-and-hold and rule-based trading?

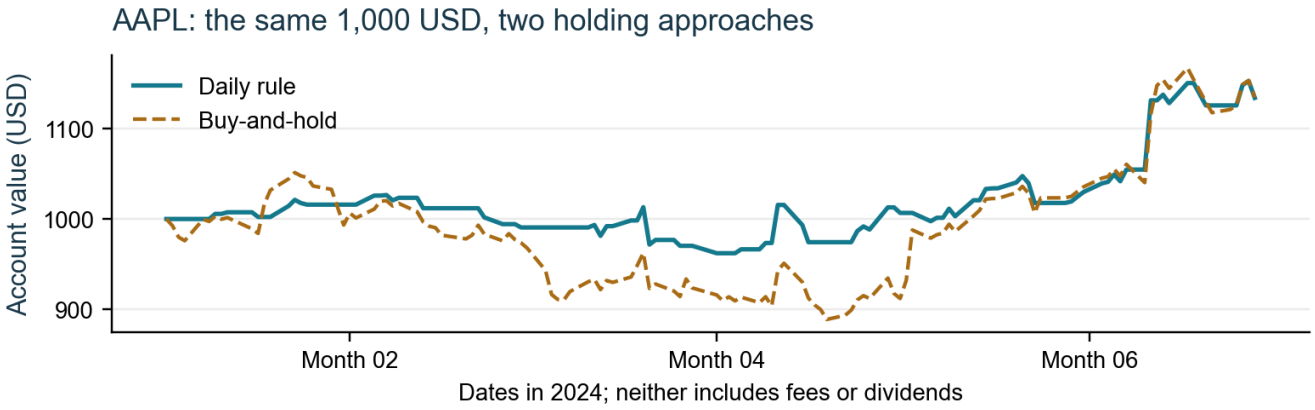
Compare two account paths using the same data, initial capital and final valuation date.

Both start with 1,000 USD. Buy-and-hold buys at the 1/2 close of 185.64; the daily rule starts in cash and trades on the earlier dates. Value both on 6/28 at 210.62 USD/share, without forced selling; neither includes dividends or fees.

Predict: Both accounts are worth 1,000 USD on 1/2. Must their 6/28 values be equal?

```
buy_hold = 1000 * close / close.iloc[0]
dates = ["2024-01-02", "2024-01-10", "2024-06-28"]
pd.DataFrame({
    'Observation date': dates,
    'Buy-and-hold (USD)': [f"{buy_hold[d]:.2f}" for d in dates],
    'Daily rule (USD)': [f"{wealth[d]:.2f}" for d in dates]})
```

Observation date	Buy-and-hold (USD)	Daily rule (USD)
2024-01-02	1000.00	1000.00
2024-01-10	1002.96	1005.67
2024-06-28	1134.56	1134.06



6/28 final values: rule 1,134.06 USD; buy-and-hold 1,134.56 USD. Final values are close, but their paths differ.

$$1000\left(\frac{210.62}{185.64}\right) \approx 1134.56 \quad \text{(USD)}$$

On 6/28, buy-and-hold has 1134.56 USD and the daily rule 1134.06 USD, about 0.50 USD less. This period's final values cannot establish equal risk or which approach will earn more in the future.

For example, cash on 1/8 misses +2.4175%, while cash on 1/11 avoids -0.3223%. Sometimes it avoids losses, sometimes it misses gains. Do not select one favorable date to represent the whole strategy. Both lines exclude fees.

Discussion

Do close final values mean the accounts had equal values throughout? Explain using the table's 1/10 amounts.

Sources: [4], [5].

ISSUE 12

Do daily and five-day price changes lead to trades on the same dates?

Different lookbacks can produce opposite requests today; both execute only at the next close.

Keep the same data and 1,000 USD. Change the complete holding condition: a daily or five-interval return >0 requests stock; otherwise, cash. This is not an entry-only change with a different exit rule. Five intervals require six closing prices.

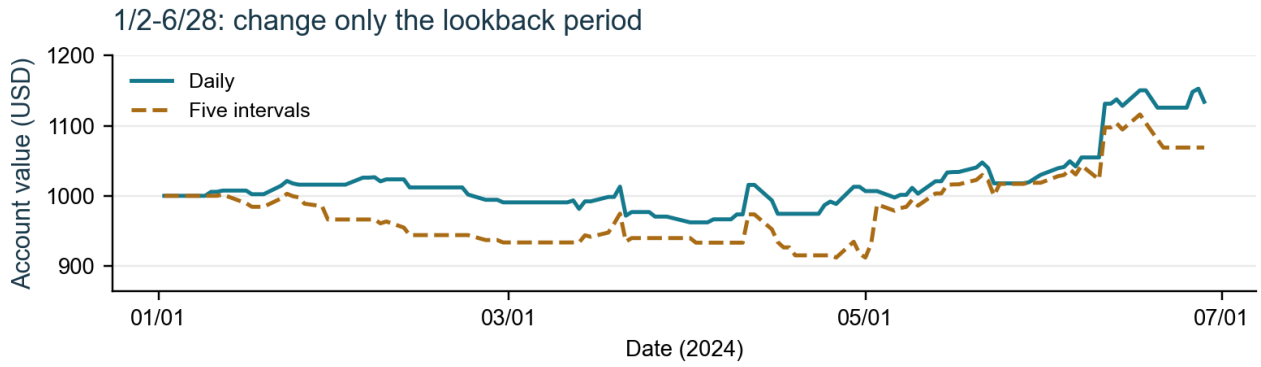
$$\frac{185.59}{186.19} - 1 \approx -0.3223\%, \qquad \frac{185.59}{181.91} - 1 \approx +2.0230\%$$

Both calculations end on 1/11: the daily calculation starts on 1/10 and the five-interval calculation on 1/4. These look back at observed prices; they do not forecast the next five days.

Predict: Will the two requests be equal on 1/11?

```
five_return = close.pct_change(5, fill_method=None)
five_request = (five_return > 0).astype(float).where(five_return.notna())
five_position = five_request.shift(2).fillna(0)
five_after = five_request.shift(1).fillna(0)
five_after.iloc[-1] = five_position.iloc[-1]  # 6/28: valuation only
five_wealth = 1000 * (1 + asset_return.where(five_position.eq(1), 0)).cumprod()
dates = ["2024-01-10", "2024-01-11", "2024-01-12"]
pd.DataFrame({'Decision date': dates,
    'Daily request': ['Stock holding' if signal[d] == 1 else 'Cash' for d in dates],
    'Five-interval request': ['Stock holding' if five_request[d] == 1 else 'Cash' for d in dates]})
```

Decision date	Daily request	Five-interval request
2024-01-10	Stock holding	Stock holding
2024-01-11	Cash	Stock holding
2024-01-12	Stock holding	Stock holding



On 6/28, the daily condition gives 1134.06 USD and the five-interval condition 1068.80 USD. This compares two complete holding conditions, not forecasting ability.

Discussion

If stock is already held on 1/11, which condition requests a sale on 1/12? Explain using the starting prices above.

Sources: [3], [4].

ISSUE 13

Which trades change if buying requires two consecutive rises?

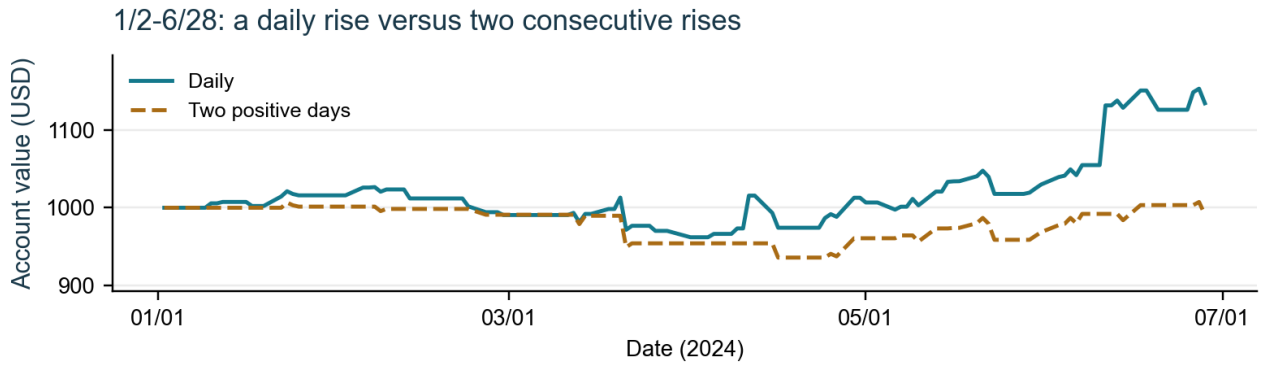
An additional confirmation condition may delay entry; it does not guarantee a further rise.

Request stock only when both adjacent daily returns are >0; otherwise, cash. Execution remains at the next close. The daily condition uses only the latest rise. 1/17 falls and 1/18 rises, so 1/18 does not meet the two-day condition. A further rise on 1/19 permits a request to buy on 1/22.

Predict: The price rises on 1/18. Does the two-day condition also request buying?

```
two_request = ((asset_return > 0) & (asset_return.shift(1) > 0))
two_request = two_request.astype(float).where(asset_return.shift(1).notna())
two_position = two_request.shift(2).fillna(0)
two_after = two_request.shift(1).fillna(0)
two_after.iloc[-1] = two_position.iloc[-1]
two_wealth = 1000 * (1 + asset_return.where(two_position.eq(1), 0)).cumprod()
dates = ["2024-01-18", "2024-01-19", "2024-01-22"]
pd.DataFrame({'Decision date': dates,
    'Current daily return (%)': (100*asset_return.loc[dates]).round(4).values,
    'Previous daily return (%)': (100*asset_return.shift(1).loc[dates]).round(4).values,
    'Two-day request': ['Stock holding' if two_request[d] else 'Cash' for d in dates]})
```

Decision date	Current daily return (%)	Previous daily return (%)	Two-day request
2024-01-18	3.2571	-0.5174	Cash
2024-01-19	1.5533	3.2571	Stock holding
2024-01-22	1.2163	1.5533	Stock holding



Final values are 1134.06 and 991.00 USD respectively. Waiting for another confirmation may avoid some declines or miss some rises.

If either of the two daily returns is not greater than 0 at the close check, request cash at the next close; an existing holding is sold then. If stock is still requested, continue holding without another purchase.

Discussion

If the buy request forms only on 1/19, can the 1/18 to 1/19 rise count as account profit? Identify the actual entry date.

Sources: [3], [4].

ISSUE 14

Hold stock after any rise, or only after a rise above 0.5%?

A stricter holding condition does not remove only losing trades.

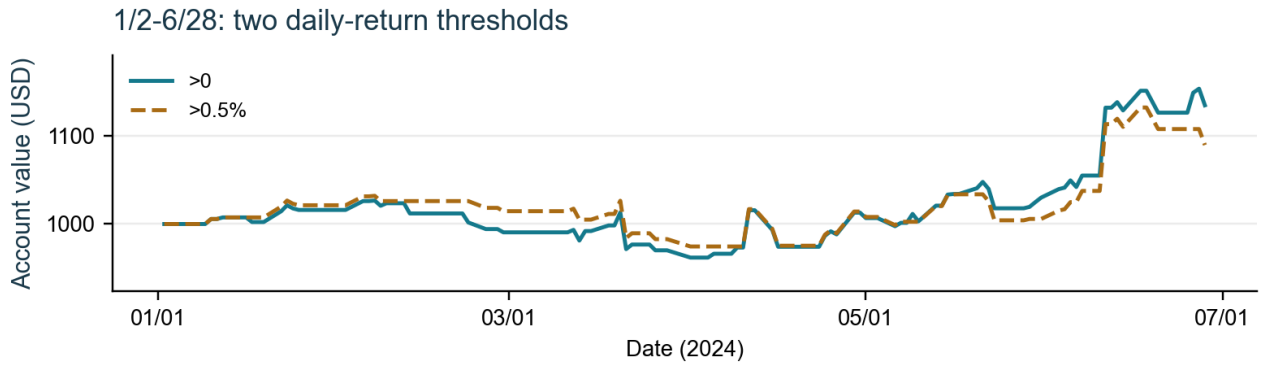
Compare two complete requests: daily return >0 or >0.5% requests stock; otherwise, cash. 0.5%=0.005. The inequality is strict, so exactly 0.5% does not qualify. Both entry and exit follow this request; the original exit condition is not retained separately.

The 1/12 close of 185.92 exceeds 185.59 on 1/11, but rises only about 0.1778%. The first condition therefore requests stock and the second cash. This is an assumed threshold, not an expected future profit.

Predict: The price rises on 1/12. Does it also meet the 0.5% condition?

```
threshold_request = (asset_return > .005).astype(float).where(asset_return.notna())
threshold_position = threshold_request.shift(2).fillna(0)
threshold_after = threshold_request.shift(1).fillna(0)
threshold_after.iloc[-1] = threshold_position.iloc[-1]
threshold_wealth = 1000 * (1 + asset_return.where(threshold_position.eq(1), 0)).cumprod()
dates = ["2024-01-10", "2024-01-11", "2024-01-12"]
pd.DataFrame({'Decision date': dates,
    'Daily return (%)': (100*asset_return.loc[dates]).round(4).values,
    '>0 request': ['Stock holding' if signal[d] == 1 else 'Cash' for d in dates],
    '>0.5% request': ['Stock holding' if threshold_request[d] == 1 else 'Cash' for d in dates]})
```

Decision date	Daily return (%)	>0 request	>0.5% request
2024-01-10	0.5671	Stock holding	Stock holding
2024-01-11	-0.3223	Cash	Cash
2024-01-12	0.1778	Stock holding	Cash



Final values are 1134.06 and 1089.21 USD. A stricter condition may avoid declines or reduce exposure to intervals that later rise.

Discussion

If stock is already held and the 1/12 rise is only 0.1778%, what does the second rule do on the next trading day, 1/16?

Sources: [3], [4].

ISSUE 15

When do you sell after holding for five fixed trading intervals?

Stop switching daily with price changes; schedule the holding period at entry.

From this issue, exit examples separately assume investing 1,000 USD at the 1/18 close, buying at 188.63 USD/share. There is no re-entry; fees and interest are 0. This is not the earlier daily rule's first purchase date. This segment illustrates different exit conditions, not the whole year or optimal timing.

The five intervals are 1/18→19→22→23→24→25, requiring six Close observations. Selling at the 1/25 close is scheduled in advance. Intermediate rises or falls do not change that date; this is not a decision made after seeing the 1/25 price to sell the next day.

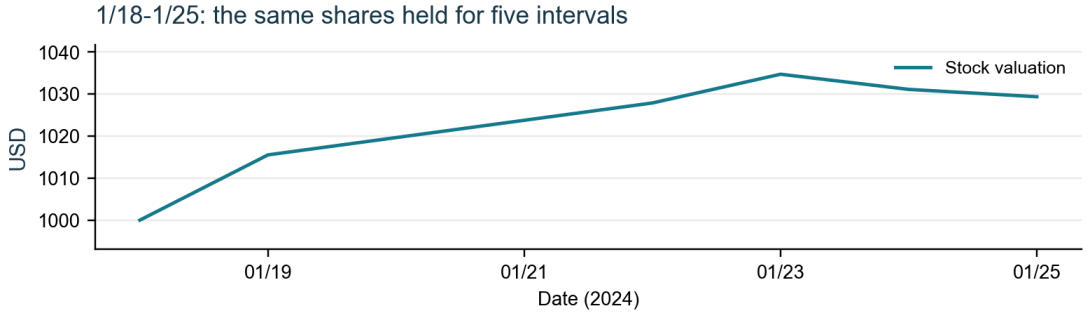
Predict: How many holding-return intervals lie between the six dates?

```
single_close = close.loc["2024-01-18":"2024-03-28"]
single_entry = single_close.iloc[0]
single_shares = 1000 / single_entry
fixed_dates = single_close.index[:6]
fixed_values = single_shares * single_close.loc[fixed_dates]
pd.DataFrame({'Observation date': fixed_dates,
    'Closing price (USD/share)': single_close.loc[fixed_dates].round(2).values,
    'Stock holding value (USD)': fixed_values.round(2).values})
```

Observation date	Closing price (USD/share)	Stock holding value (USD)
2024-01-18	188.63	1000.0
2024-01-19	191.56	1015.53
2024-01-22	193.89	1027.89
2024-01-23	195.18	1034.72
2024-01-24	194.5	1031.12
2024-01-25	194.17	1029.37

q is the number of shares bought; W is total stock and cash value (USD). The share quantity is fixed, so before selling W equals shares times the closing price.

$$q = 1000/188.63 \approx 5.30138, \quad W_{1/25} \approx 5.30138(194.17) \approx 1029.37 \text{ USD}$$



The assumed 1/25 sale leaves 1029.37 USD. Later take-profit and stop-loss examples change the exit condition while keeping the entry price and 1,000 USD fixed.

Discussion

If a rise above 3% leads to early selling on 1/23, is it still this issue's fixed five-interval rule?

Sources: [2], [4].

ISSUE 16

If the close is 2% below entry, is the actual loss only 2%?

A stop-loss condition determines when to request an exit; execution prices still determine actual profit and loss.

Again buy on 1/18 at 188.63 USD/share with 1,000 USD, observing through 3/28. Check each close; a fall of at least 2% requests selling at the next close. After selling, cash earns no interest and there is no re-entry. This is a closing-price rule, not a broker's intraday stop order. [12]

C_{buy} is the entry price and C_{stop} the closing-price trigger level, both in USD/share.

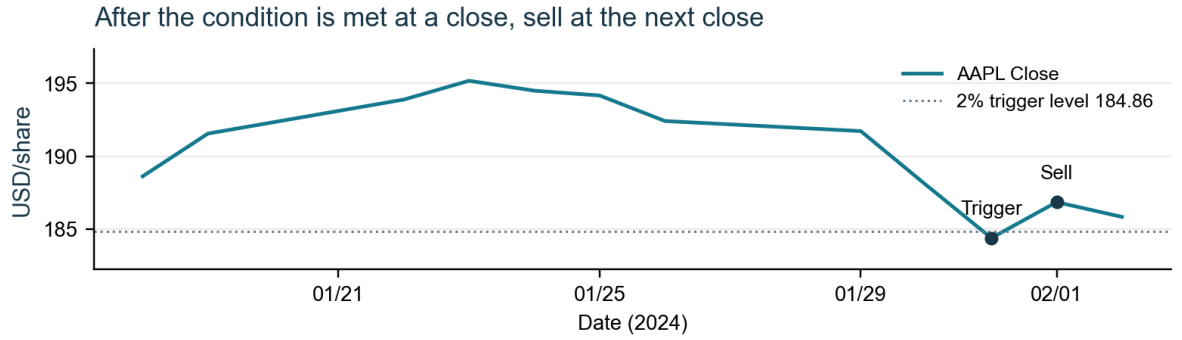
$$C_{stop} = C_{buy}(1 - 0.02) = 188.63(0.98) \approx 184.8574 \quad \text{(USD/share)}$$

Predict: Will the condition be met and the sale occur on the same day?

```
loss_level = single_entry * .98
loss_hit = int(np.flatnonzero(single_close.values <= loss_level)[0])
loss_sale = loss_hit + 1
assert loss_sale < len(single_close)-1  # This example has a next close before 3/28
loss_trigger = single_close.index[loss_hit]
loss_sell = single_close.index[loss_sale]
loss_terminal = single_shares * single_close.iloc[loss_sale]
dates = [single_close.index[0], loss_trigger, loss_sell]
pd.DataFrame({'Observation date': dates, 'Event': ['Buy', 'Condition met', 'Assumed sale'],
    'Closing price (USD/share)': single_close.loc[dates].round(2).values})
```

Observation date	Event	Closing price (USD/share)
2024-01-18	Buy	188.63
2024-01-31	Condition met	184.4
2024-02-01	Assumed sale	186.86

The 1/31 close of 184.40 is below 184.8574, but 2/1 rebounds to 186.86. The sale leaves 990.62 USD, a loss of 9.38 USD or about 0.9383%, not exactly 2%.



The dashed line is a trigger condition, not an execution price. The markers separate the 1/31 trigger from the 2/1 sale. If the next day continues falling, actual loss may exceed 2%.

Discussion

Buying at 188.63 and selling at 186.86, how much of 1,000 USD remains? Use the actual entry price as the denominator.

Sources: [2], [12].

ISSUE 17

With a 2% or 4% stop loss, when do you sell and how much remains?

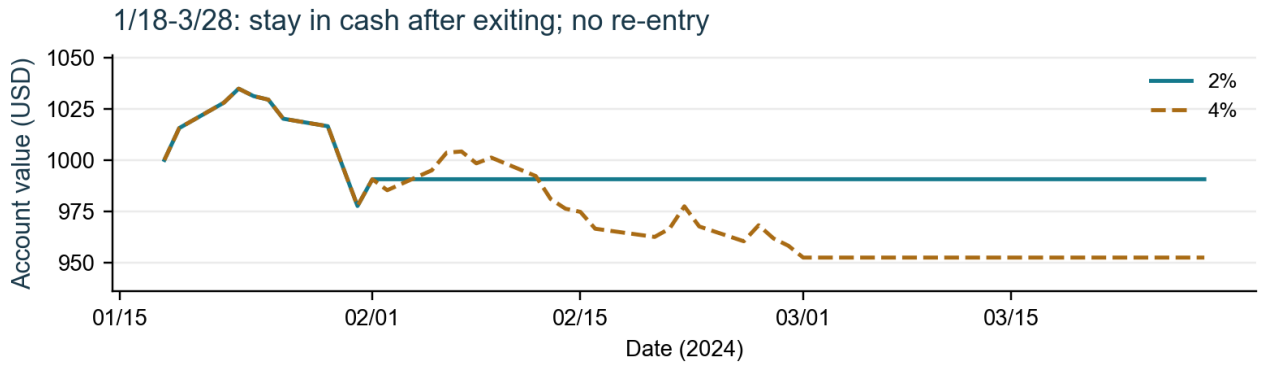
Change only the permitted decline relative to entry; entry, close-based checks, next-close execution and common valuation date stay fixed.

Again buy 1,000 USD on 1/18 at 188.63 USD/share. Do not buy again after selling; cash earns no interest. Compare total values on 3/28. Test both thresholds using full-precision prices, without first rounding prices or returns.

Predict: If the stop loss widens from 2% to 4%, will this 1,000 USD trade lose exactly 20 USD more?

```
loss_rows, loss_paths = [], {}
for cut in [.02, .04]:
    level = single_entry * (1-cut)
    hit = int(np.flatnonzero(single_close.values <= level)[0])
    sale = hit + 1
    assert sale < len(single_close)-1
    value = single_shares * single_close.iloc[sale]
    path = single_shares * single_close.copy()
    path.iloc[sale:] = value  # All cash after selling; retain the common 3/28 endpoint
    loss_paths[f"{cut:.0%}"] = path
    loss_rows.append([f"{cut:.0%}", single_close.index[hit],
        single_close.index[sale], round(value, 2)])
pd.DataFrame(loss_rows, columns=['Stop-loss percentage', 'Trigger date', 'Sale date', '3/28 total value (USD)'])
```

Stop-loss percentage	Trigger date	Sale date	3/28 total value (USD)
2%	2024-01-31	2024-02-01	990.62
4%	2024-02-29	2024-03-01	952.45



The 2% condition leaves about 990.62 USD after selling on 2/1; the 4% condition leaves about 952.45 USD after selling on 3/1. A flat line means cash, not an unchanged stock price.

The 4% condition triggers on 2/29, but 3/1 falls further to 179.66 USD/share, producing an actual loss of about 4.7553%. This segment does not establish a universally optimal threshold.

Discussion

The 4% condition actually sells at 179.66. What percentage does the trade lose? Use the 188.63 entry price as the denominator and explain why it is not 4%.

Sources: [2], [12].

ISSUE 18

How do you take profit when the close is 3% above entry?

Meeting the profit condition requests a sale; 3% is not directly a realized return.

Again buy on 1/18 at 188.63 USD/share with 1,000 USD. A close at least 3% above entry leads to selling at the next close. There is no re-entry or cash interest; compare on 3/28. Only a take-profit condition is set, with no exit condition for a decline.

C_{buy} is the entry price and C_{target} the take-profit trigger level, both in USD/share.

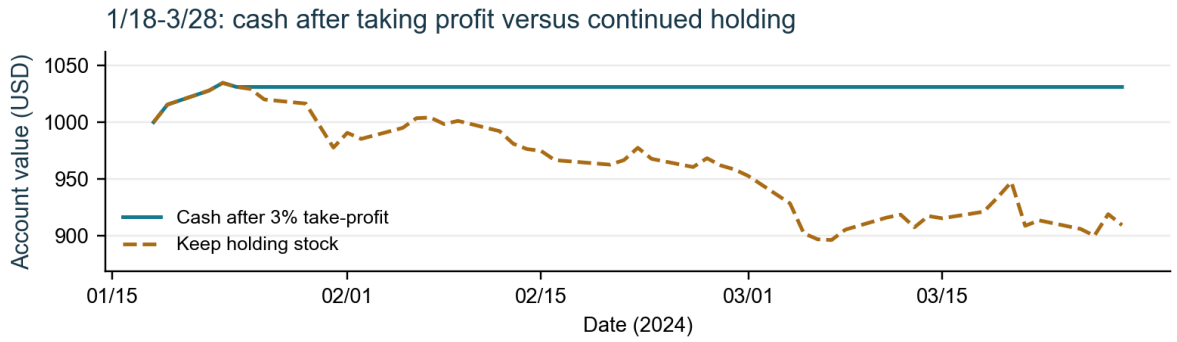
$$C_{target} = C_{buy}(1 + 0.03) = 188.63(1.03) \approx 194.2889 \quad \text{(USD/share)}$$

Predict: Which is this example's sale price: 195.18 on the trigger day or 194.50 at the next close?

```
profit_level = single_entry * 1.03
profit_hit = int(np.flatnonzero(single_close.values >= profit_level)[0])
profit_sale = profit_hit + 1
assert profit_sale < len(single_close)-1
profit_trigger, profit_sell = single_close.index[[profit_hit, profit_sale]]
profit_terminal = single_shares * single_close.iloc[profit_sale]
profit_path = single_shares * single_close.copy()
profit_path.iloc[profit_sale:] = profit_terminal
dates = [single_close.index[0], profit_trigger, profit_sell]
pd.DataFrame({'Observation date': dates, 'Event': ['Buy', 'Condition met', 'Assumed sale'],
    'Closing price (USD/share)': single_close.loc[dates].round(2).values})
```

Observation date	Event	Closing price (USD/share)
2024-01-18	Buy	188.63
2024-01-23	Condition met	195.18
2024-01-24	Assumed sale	194.5

The condition is met on 1/23; an assumed sale on 1/24 at 194.50 leaves 1031.12 USD, earning about 3.1119%. Shares are still calculated as 1000÷188.63; do not substitute 194.2889 as a new entry price.



At the common 3/28 endpoint, cash is about 1,031.12 USD and the continued holding about 909.08 USD. The gap comes from the observed price path after exit, not a guarantee of a more favorable result next time.

Discussion

If the 1/23 price of 195.18 is wrongly used instead of the 1/24 sale price of 194.50, how much is profit overstated on this 1,000 USD trade?

Sources: [2], [12].

ISSUE 19

Could a 3% or 5% take-profit condition remain unmet?

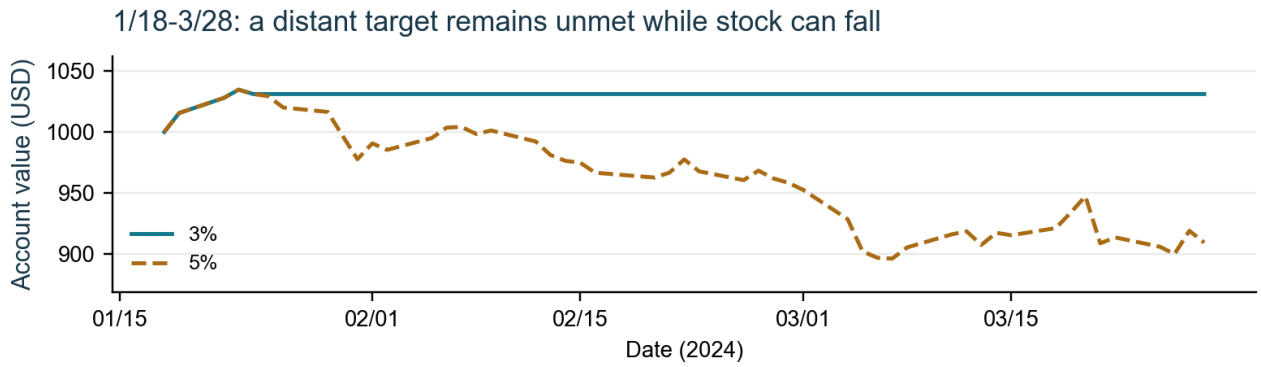
A more distant profit target is not guaranteed; with only take-profit, stock may remain held while prices fall.

Again buy on 1/18 at 188.63 USD/share with 1,000 USD; change the profit condition. 3% requires a close of at least 194.2889, and 5% requires 198.0615 USD/share. Value unsold stock on 3/28 at 171.48; do not treat it as 0 or invent a sale.

Predict: Does the higher 5% condition guarantee more money at the end?

```
profit_rows, profit_paths = [], {}
for gain in [.03, .05]:
    level = single_entry * (1+gain)
    hits = np.flatnonzero(single_close.values >= level)
    path = single_shares * single_close.copy()
    if len(hits) and hits[0]+1 < len(single_close)-1:
        hit, sale = int(hits[0]), int(hits[0])+1
        path.iloc[sale:] = path.iloc[sale]  # Hold cash after selling; no re-entry
        trigger, end_action = single_close.index[hit], single_close.index[sale]
    else:
        trigger, end_action = 'Not reached', '3/28 stock valuation'
    profit_paths[f"{gain:.0%}"] = path
    profit_rows.append([f"{gain:.0%}", round(level, 4), trigger,
        end_action, round(path.iloc[-1], 2)])
pd.DataFrame(profit_rows, columns=['Take-profit percentage', 'Trigger level',
    'Trigger date', 'Sale or valuation', '3/28 USD'])
```

Take-profit percentage	Trigger level	Trigger date	Sale or valuation	3/28 USD
3%	194.2889	2024-01-23	2024-01-24	1031.12
5%	198.0615	Not reached	3/28 stock valuation	909.08



This segment's highest Close is about 195.18, below 198.0615. The 5% rule keeps holding and is worth 909.08 USD on 3/28. The 3% rule has exited, leaving 1,031.12 USD cash.

Discussion

Does setting only a 5% take-profit condition prevent losses? Calculate the change in this 1,000 USD investment using entry at 188.63 and the 3/28 price of 171.48.

Sources: [2], [12].

ISSUE 20

Does a trailing stop protect profit after the stock has risen?

The exit level rises with observed highs, but delayed execution and price gaps can still produce a sale below entry.

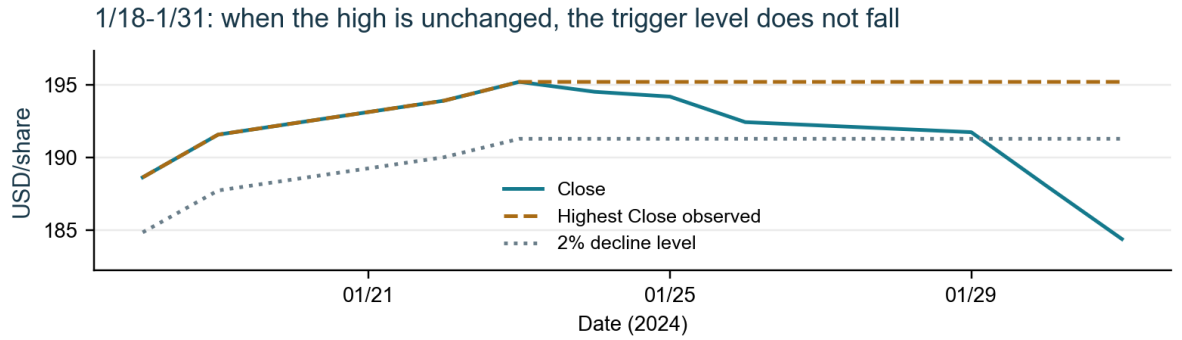
Again buy on 1/18 at 188.63 USD/share. A fixed stop loss uses entry; this issue uses the highest close observed since entry. C_{max} is the highest Close observed through today, and C_{stop} the trigger level 2% below it (USD/share). Check closes and sell at the next close; this is not a broker's intraday trailing-stop order. [12]

$$C_{stop} = 0.98C_{max}, \qquad 0.98(195.18) \approx 191.2764 > 188.63$$

Predict: If the highest Close no longer rises, does the trigger level fall with today's price?

```
peak_close = single_close.cummax()  # Each row uses only Close observations through that day
trail_level = .98 * peak_close
trail_hit = int(np.flatnonzero(single_close.values <= trail_level.values)[0])
trail_sale = trail_hit + 1
assert trail_sale < len(single_close)-1
trail_terminal = single_shares * single_close.iloc[trail_sale]
dates = ["2024-01-23", "2024-01-29", "2024-01-30", "2024-01-31"]
pd.DataFrame({'Observation date': dates,
    'Closing price': single_close.loc[dates].round(2).values,
    'Highest Close': peak_close.loc[dates].round(2).values,
    'Trigger price': trail_level.loc[dates].round(4).values})
```

Observation date	Closing price	Highest Close	Trigger price
2024-01-23	195.18	195.18	191.2764
2024-01-29	191.73	195.18	191.2764
2024-01-30	188.04	195.18	191.2764
2024-01-31	184.4	195.18	191.2764



The 1/23 high of 195.18 raises the level to 191.2764. The 1/30 close of 188.04 is below it, so sell on 1/31 at 184.40. This leaves 977.58 USD, still a loss.

Discussion

Why is there a final loss even though the trigger level of 191.2764 exceeds entry at 188.63? Identify the trigger date and actual sale price.

Sources: [2], [12].

ISSUE 21

What changes when a 2% stop loss is added to a 5% take-profit condition?

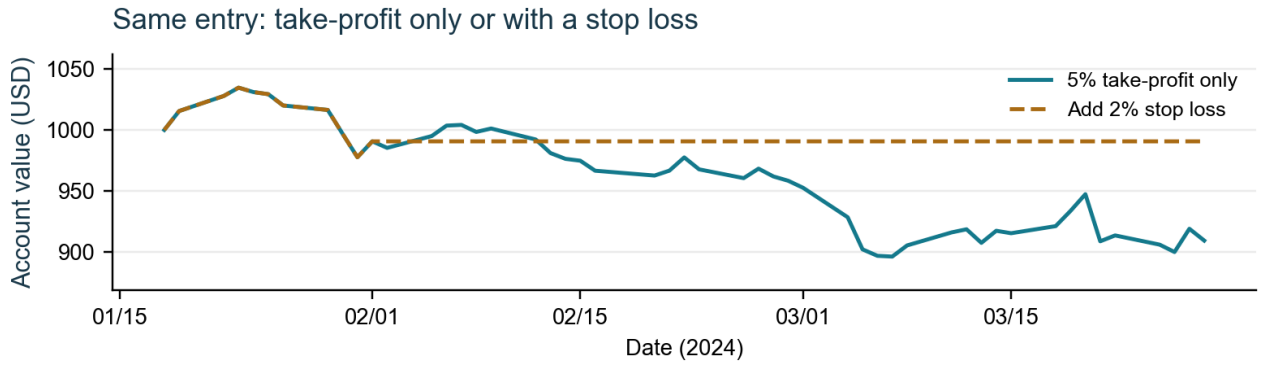
Keep the profit target and add an exit condition for a price decline.

Both invest 1,000 USD on 1/18 at 188.63 USD/share and compare values on 3/28. The take-profit level is 198.0615 and the stop-loss level 184.8574 USD/share. Check each close: whichever condition is met first leads to selling at the next close. No re-entry or cash interest follows.

Predict: The 5% take-profit condition is never met in this segment. After adding a stop loss, is stock still held through 3/28?

```
combined_hits = np.flatnonzero(
    (single_close.values <= single_entry*.98) |
    (single_close.values >= single_entry*1.05))
combined_hit = int(combined_hits[0])
combined_sale = combined_hit + 1
assert combined_sale < len(single_close)-1
combined_terminal = single_shares * single_close.iloc[combined_sale]
combined_path = single_shares * single_close.copy()
combined_path.iloc[combined_sale:] = combined_terminal
dates = [single_close.index[combined_hit], single_close.index[combined_sale]]
pd.DataFrame({'Observation date': dates, 'Event': ['Stop-loss condition met', 'Sell at next close'],
    'Closing price (USD/share)': single_close.loc[dates].round(2).values})
```

Observation date	Event	Closing price (USD/share)
2024-01-31	Stop-loss condition met	184.4
2024-02-01	Sell at next close	186.86



The stop loss is met on 1/31; selling on 2/1 leaves 990.62 USD. Take-profit only still holds stock on 3/28, worth 909.08 USD.

In this segment, the price first falls to the stop-loss condition. Adding it prevents subsequent declines from affecting the account. If prices recover after exit, cash also misses that recovery. One segment cannot establish that stop losses necessarily increase profit.

Discussion

At the common 3/28 endpoint, how many more USD does the account have after adding the stop loss? Use the two final values, not 2% as the actual loss.

Sources: [2], [12].

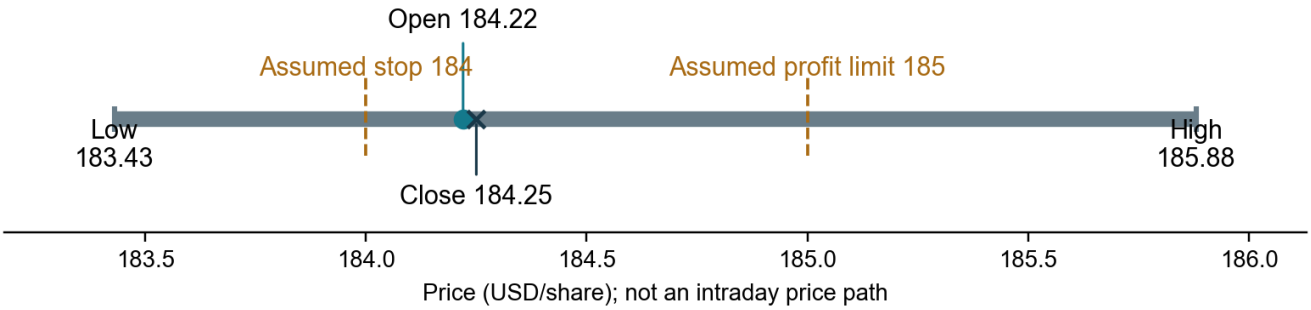
ISSUE 22

If stop and profit levels are both touched in one day, which occurs first?

The high and low give a range, not an event sequence.

Separately examine 1/3: assume buying one share at the open with both stop-loss and take-profit orders, cancelling the other order after one sale executes. These assumed orders do not change the main account. The 184 USD stop is a trigger: it generally becomes a market order when reached, without guaranteed execution at 184. The 185 USD profit level is a limit: execution must be at 185 or higher, and a touch may not fill. [2]

AAPL 2024/1/3: the range has no intraday sequence



Both 184 and 185 lie between 183.43 and 185.88; this figure cannot establish which stop-loss or take-profit event occurs first.

Prices are in USD/share. The open is 184.22; the low of 183.43 is below 184 and the high of 185.88 above 185. Both levels are within the day's range. The 184.25 close does not tell us whether an intraday rise or fall came first.

A fall to 184 first may trigger the stop first; a rise to 185 first may fill the take-profit order first. Both sequences fit the daily range. Both rows below only assume execution at the stated levels; they are not actual execution prices.

Predict: If you always select an exit at 185 USD, are you always choosing the more favorable assumption?

```
entry = aapl.loc["2024-01-03", "Open"]
exit_assumptions = [184.0, 185.0]
pd.DataFrame({
    'Assumed exit': ['Fill at stop level', 'Fill at profit level'],
    'Price (USD/share)': exit_assumptions,
    'Profit/loss per share (USD)': [f"{c-entry:+.2f}" for c in exit_assumptions],
    'Return': [f"{c/entry-1:+.4%}" for c in exit_assumptions]})
```

Assumed exit	Price (USD/share)	Profit/loss per share (USD)	Return
Fill at stop level	184.0	-0.22	-0.1194%
Fill at profit level	185.0	+0.78	+0.4234%

$$\frac{185}{184.22} - 1 \approx +0.4234\%$$

The assumptions differ by 1 USD per share: a loss of 0.22 or profit of 0.78 USD. Neither is an observed trade or a bound on actual profit/loss. Finer price data may help identify sequence, but cannot replace actual execution records.

Discussion

Can this day's OHLC alone establish that the trade must earn 0.78 USD per share?

Sources: [2].

ISSUE 23

Which trades are skipped when waiting three closes after a sale?

Keep the exit condition; restrict only how soon another buy decision can be made after selling.

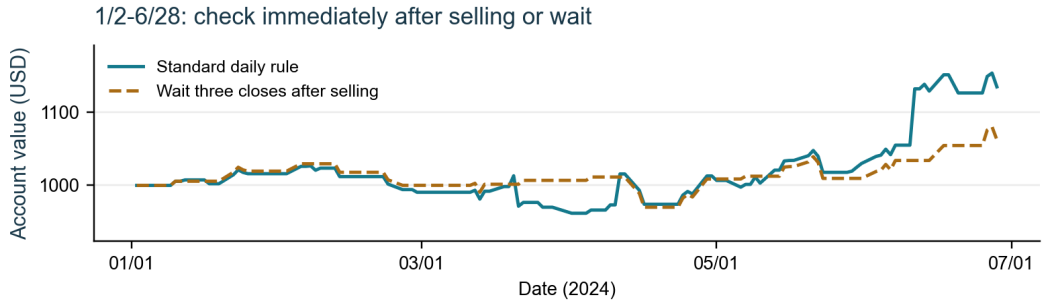
Use the full-period daily rule, next-close execution and no cash interest. After a 1/10 sale, make no buy decision that day or at the next three trading-day closes (1/11, 12, 16); checks resume on 1/17. Rises during those days cannot permit an early purchase.

Predict: The price rises on 1/12. Can you buy again yet? When does the 1/18 buy decision execute?

```
wait_request = pd.Series(0.0, index=close.index)
last_sale = -100  # No previous sale at the start, so no waiting is needed
for j in range(len(close)):
    held = wait_request.iloc[j-1] if j >= 1 else 0
    held_before = wait_request.iloc[j-2] if j >= 2 else 0
    if held_before == 1 and held == 0:
        last_sale = j  # Sale actually executes at today's close; count the wait from today
    allowed = held == 1 or j-last_sale >= 4
    wait_request.iloc[j] = float(allowed and asset_return.iloc[j] > 0)
wait_position = wait_request.shift(2).fillna(0)
wait_after = wait_request.shift(1).fillna(0)
wait_after.iloc[-1] = wait_position.iloc[-1]  # 6/28: valuation only
wait_wealth = 1000*(1+asset_return.where(wait_position.eq(1), 0)).cumprod()
dates = ["2024-01-12", "2024-01-17", "2024-01-18", "2024-01-19"]
pd.DataFrame({'Observation date': dates, 'Daily stock return': asset_return.loc[dates].map('{:+.4%}'.format),
    'After-close stock request': wait_request.loc[dates].astype(int).values,
    'Stock held after close execution': wait_after.loc[dates].astype(int).values})
```

Observation date	Daily stock return	After-close stock request	Stock held after close execution
2024-01-12	+0.1778%	0	0
2024-01-17	-0.5174%	0	0
2024-01-18	+3.2571%	1	0
2024-01-19	+1.5533%	1	1

Despite the 0.1778% rise on 1/12, the wait continues. Checks resume on 1/17, but the price falls. The 1/18 rise requests buying at the 1/19 close. Waiting three days is an additional rule, not a forecast that prices must rise afterward.



With the same prices, the standard rule ends at 1134.06 USD and the waiting rule at 1062.06 USD. Fees are not yet included on this page.

Discussion

Under the waiting rule, can the 1/18 to 1/19 rise count as a gain on stock already held? Identify the actual entry time.

Sources: [2], [4].

ISSUE 24

Do fewer trades necessarily leave more money at the end?

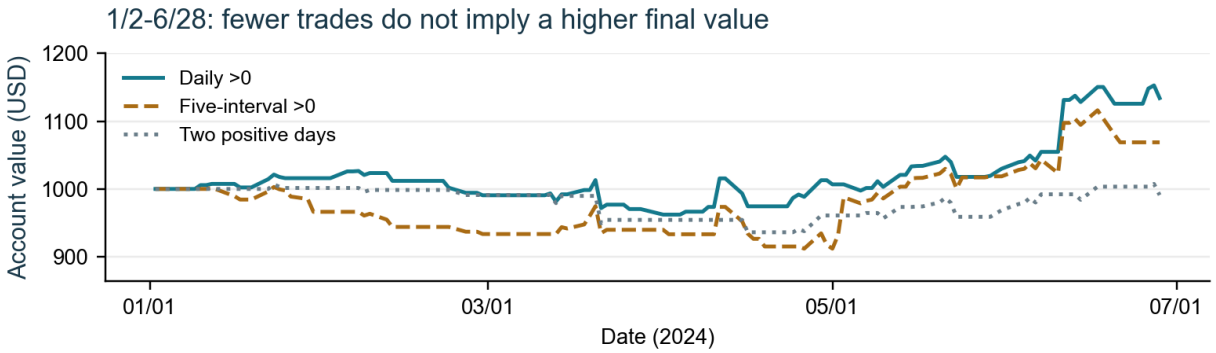
Compare complete rules over the same period; trade counts and returns are different outcomes.

All use 1/2-6/28, 1,000 USD, next-close execution, fractional shares, no cash interest and, for now, no fees. The five rules decide daily between stock and cash. The single-entry 1/18 take-profit/stop-loss examples use a different period and are excluded from this table.

Predict: The two-consecutive-rises rule trades less. Must its final value exceed the daily rule's?

```
rule_positions = {'Daily >0': position, 'Five-interval >0': five_position,
    'Two positive days': two_position, 'Daily rise >0.5%': threshold_position,
    'Wait 3 closes after selling': wait_position}
rule_after = {'Daily >0': after_close, 'Five-interval >0': five_after,
    'Two positive days': two_after, 'Daily rise >0.5%': threshold_after,
    'Wait 3 closes after selling': wait_after}
rule_wealth = {'Daily >0': wealth, 'Five-interval >0': five_wealth,
    'Two positive days': two_wealth, 'Daily rise >0.5%': threshold_wealth,
    'Wait 3 closes after selling': wait_wealth}
comparison = []
for name in rule_positions:
    moves = rule_after[name] - rule_positions[name]
    comparison.append([name, int((moves>0).sum()), int((moves<0).sum()),
                       round(rule_wealth[name].iloc[-1], 2)])
pd.DataFrame(comparison, columns=['Rule', 'Purchases', 'Sales', '6/28 USD'])
```

Rule	Purchases	Sales	6/28 USD
Daily >0	28	27	1134.06
Five-interval >0	15	15	1068.8
Two positive days	13	12	991.0
Daily rise >0.5%	27	26	1089.21
Wait 3 closes after selling	15	14	1062.06



The two-positive-days rule buys 13 times and sells 12, ending at 991.00 USD. The daily rule buys 28 times and sells 27, ending at 1134.06 USD.

Count actual purchases and sales; continuing a holding is not another purchase. No trades execute on 6/28, so some rules end with stock. These results apply to this data segment. If a rule is chosen after seeing the 2024 results, validate it on another period not used for selection.

Discussion

How many fewer actual trades does the two-positive-days rule make than the daily rule, and how many fewer USD does it end with? Do those gaps prove that fewer trades are necessarily worse?

Sources: [4], [5].

ISSUE 25

Does charging 1 USD per purchase or sale change the ranking?

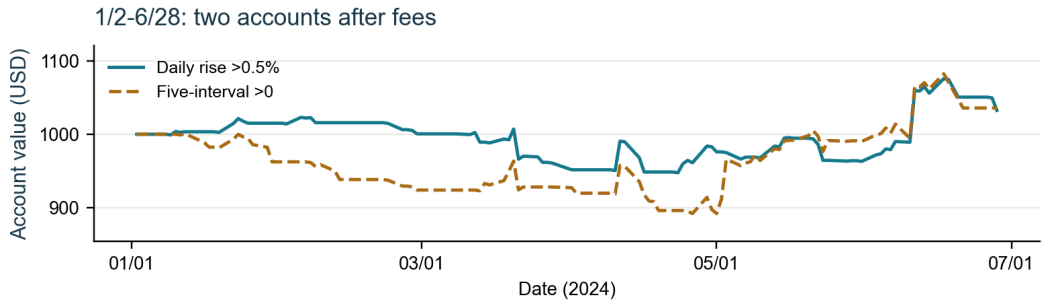
Keep the five rules and execution dates fixed; add only a deduction at execution.

Assume 1 USD per purchase or sale, with no fee for continuing a holding. Deduct the fee before buying shares and after selling into cash. 6/28 is valuation only, with no extra fee. The daily rule first buys on 1/9: 999÷185.14≈5.39592 shares. All amounts below are in USD.

Predict: The daily-rise >0.5% rule has more money than the five-interval rule before fees. Is that still true after fees?

```
cost_results, cost_rows = {}, []
for name, targets in rule_after.items():
    cash, units, paid, balances = 1000.0, 0.0, 0, []
    for date, price in close.items():
        if targets[date] == 1 and units == 0:
            assert cash > 1
            units, cash = (cash-1)/price, 0.0
            paid += 1
        elif targets[date] == 0 and units > 0:
            cash, units = units*price-1, 0.0
            paid += 1
        balances.append(cash+units*price)
    path = pd.Series(balances, index=close.index)
    cost_results[name] = {"wealth": path, "fees": paid}
    cost_rows.append([name, paid, round(rule_wealth[name].iloc[-1], 2),
                      round(path.iloc[-1], 2)])
pd.DataFrame(cost_rows, columns=['Rule', 'Total fees', 'Final value without fees', 'Final value after fees'])
```

Rule	Total fees	Final value without fees	Final value after fees
Daily >0	55	1134.06	1072.52
Five-interval >0	30	1068.8	1035.83
Two positive days	25	991.0	965.65
Daily rise >0.5%	53	1089.21	1032.65
Wait 3 closes after selling	29	1062.06	1031.63



Without fees: 1,089.21>1,068.80; after fees: 1,032.65<1,035.83. Fees reduce shares bought and change subsequent profit and loss.

Discussion

The daily rule pays 55 USD in fees. Why does its final value fall from 1,134.06 to 1,072.52, a decrease of about 61.54 rather than 55 USD?

Sources: [5].

ISSUE 26

How many shares for a planned 20 USD stop loss?

Calculate shares from a planned loss and loss per share. The planned loss is not guaranteed.

Use the single-entry example: buy on 1/18 at 188.63 USD/share and value on 3/28. Start with 1,000 USD, ideal fractional shares, no interest on remaining cash, and no fees, dividends or tax. A close at or below the set level leads to selling all shares at the next trading day's close, with no re-entry.

q is shares and C_{buy} the entry price (USD/share). If selling exactly at the stop level, shares×loss per share=20 USD. But the purchase cannot exceed 1,000 USD. min takes the smaller of the two share quantities.

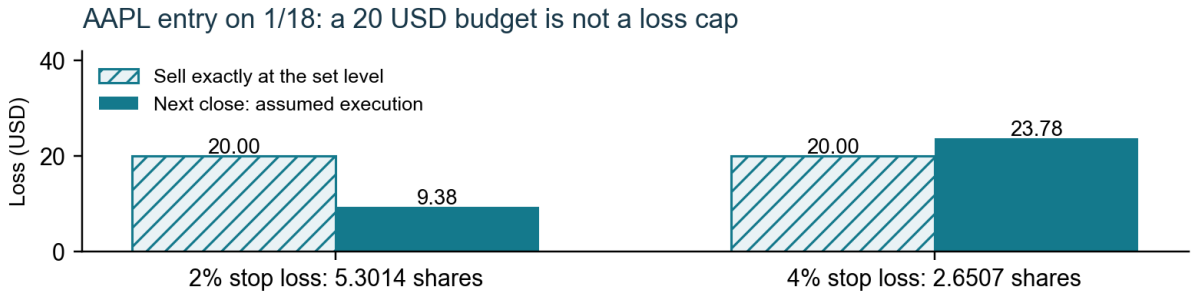
$$q = \min\left(\frac{1000}{C_{buy}}, \frac{20}{C_{buy} - \text{stop}}\right)$$

2% stop level≈184.8574: 20÷(188.63-184.8574)≈5.3014 shares, buying 1,000 USD. 4% stop level≈181.0848: 20÷7.5452≈2.6507 shares, buying 500 USD and retaining 500 USD. stop is a set level, not an execution price already obtained.

Predict: Both cases calculate shares for a planned loss of 20 USD. Will each actually lose 20 USD after selling?

```
budget_rows = []
for i, rate in enumerate([0.02, 0.04]):
    trigger = single_entry * (1-rate)
    units = min(1000/single_entry, 20/(single_entry-trigger))
    cash = 0.0 if units == 1000/single_entry else 1000-units*single_entry
    sale_date = loss_rows[i][2]  # Next-close sale date already found in Issue 17
    sale_price = single_close[sale_date]
    terminal = cash+units*sale_price
    budget_rows.append([f"{rate:.0%}", f"{units:.4f}", f"{cash:.2f}",
                        sale_date[5:], f"{sale_price:.2f}", f"{1000-terminal:.2f}"])
pd.DataFrame(budget_rows, columns=['Stop loss', 'Shares', 'Remaining cash',
    'Sale day', 'Sale price', 'Actual loss'])
```

Stop loss	Shares	Remaining cash	Sale day	Sale price	Actual loss
2%	5.3014	0.00	02-01	186.86	9.38
4%	2.6507	500.00	03-01	179.66	23.78



Amounts are in USD and sale prices in USD/share. The 2% case actually loses 9.38 and the 4% case 23.78. The next close may be above or below the set level. Both share quantity, stop-loss percentage and sale date differ; do not attribute the result only to the percentage.

Discussion

With a 1% stop loss, the 20 USD plan requires investing 2,000 USD. With only 1,000 USD and no borrowing, how many shares can you buy at most? How much would you lose if selling exactly at the set level?

Sources: [11], [12].

References

Price sources and reproduction conditions

[1] yfinance: historical prices

[2] Investor.gov: orders and execution

[3] pandas: shifting rows

[4] pandas: fractional price changes

[5] pandas: cumulative products

[6] FINRA: extended-hours trading

[7] Investor.gov: closing price

[8] Yahoo Finance: AAPL historical prices

[9] CFPB: accruing and paying deposit interest

[10] Investor.gov: interest on uninvested cash

[11] Investor.gov: fractional shares and broker restrictions

[12] Investor.gov: stop, stop-limit and trailing-stop orders

AAPL daily data: 2024/1/2-6/28, 124 observations; the Yahoo/yfinance snapshot saved on 2026/9/14 is retained. Trading rules, execution and fees are course assumptions, not actual trade records.

Prices display to 2 decimal places; calculations retain source precision. Reproduction uses the data recorded for this version, without downloading replacements.

Python 3.12.14；numpy 2.2.6；pandas 2.2.3；matplotlib 3.10.8；yfinance 1.2.0。